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Research Matching Chatbot: Bridging Students and Faculty

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Streamlining Academic Connections Through Intelligent Conversation



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Project Overview & Team

Project Snapshot & Objectives

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Chatbot Overview & Purpose

Terminal-based chatbot designed for academic environments, helping connect students and faculty by streamlining discovery of relevant research opportunities and improving communication across departments.



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Goals, Scope & Target Users

Uses AI to align student interests with faculty expertise, supporting matching, trend analysis, collaboration detection, and research gap identification for students, faculty, and academic coordinators.



Why This Project Matters



Pain Points in Faculty Discovery

Students struggle to navigate fragmented faculty information, relying on manual keyword searches that are time-consuming, inconsistent, and often fail to reveal relevant mentors or research opportunities.

Strategic Impact on Research Collaboration

Professors miss potential collaborations, while departments lack data-driven insight into emerging research trends, limiting strategic planning and coordinated, high-impact research direction setting.

Team Details

Core Team Profile

Team: M. Sai Siri Shriya and B. Akhil Rao, final-year CSE students at Vardhaman College of Engineering. They are collaborating on the Research Matching Chatbot project.



Roles and Responsibilities

Responsibilities: problem analysis, system architecture design, implementation of chatbot features, integration with datasets, and rigorous testing to ensure reliability and user-centric performance.

Problem Statement & Use Cases

Challenges for Students and Faculty

Students struggle to identify faculty whose research aligns with their interests, while professors find it difficult to track emerging trends and discover suitable collaborators across multiple domains.

Use Cases and Expected Outcomes

Supports thesis supervision, research projects, internships, and departmental expertise mapping, leading to faster, higher quality matches, better resource allocation, and more informed academic decision-making overall.

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Solution Design & Workflow

System Architecture Overview

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End-to-End Workflow & Core Components

User queries flow through retrieval, analysis, and response stages. LangChain orchestrates ChromaDB vector search and Tavily web queries to support accurate, context-rich chatbot interactions.

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Modular Design & Interaction Layers

Embeddings, vector store, and RAG pipeline collaborate while keeping matching, trend analysis, and recommendation logic separate. A terminal interface provides the primary, streamlined user interaction layer.



Matching & Recommendation Workflow



Student Input & Semantic Search Pipeline

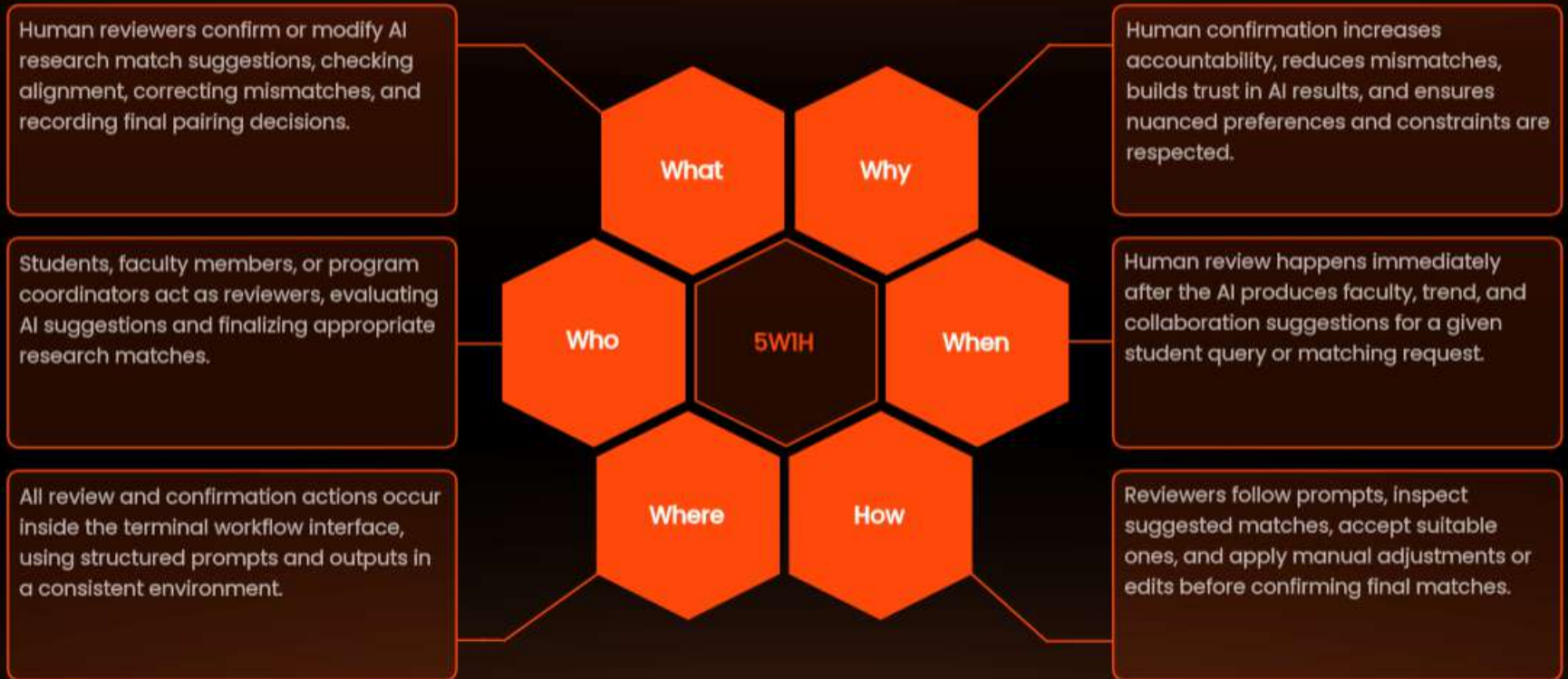
Student submits research interests in natural language; system generates semantic embeddings, searches ChromaDB faculty vector store, and retrieves top-k relevant faculty with aligned research areas.



Trend Analysis & Human-in-the-Loop Recommendations

Workflow enriches matches using Tavily-based trend analysis, surfaces collaboration opportunities and research gaps, then presents recommendations for faculty-student review and confirmation.

Human Review



Key Features Breakdown



Intelligent Matching & Retrieval

Semantic faculty matching via vector embeddings, RAG-based retrieval over profiles, and ChromaDB-powered similarity search enabling fast, scalable, context-aware discovery of relevant advisors and projects.



Research Insights & User Experience

Live trend analysis through Tavily Search API, collaboration and gap recommendations, human-in-the-loop validation, and a lightweight terminal interface supporting transparent, controllable research matching workflows.



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Technologies, Data & Impact

Technology Stack & Integration



Core technologies and orchestration

Python powers the chatbot backend, while LangChain orchestrates RAG pipelines and tool chains, enabling modular, maintainable workflows tailored to research matching scenarios.

Data storage, encoding, and secure access

ChromaDB stores and retrieves faculty vectors, HuggingFace embeddings provide semantic encoding, Tavily supplies live research trends, and dotenv secures environment variables and API keys.

Data Flow & RAG Pipeline

Step1

End-to-End Data & Retrieval Process

Students submit research statements and preferences, transformed into MiniLM embeddings. ChromaDB stores indexed faculty vectors, enabling similarity search to surface the most relevant faculty profiles efficiently.

Step2

Context Enrichment & Recommendation Output

Tavily Search augments retrieved profiles with current trends. The RAG model generates explanations, matches, collaboration ideas, and gaps, producing structured recommendations for review and logging.

Future Roadmap

Key work plan for the first half of the year

Jan. Jun.

Work Plan

Key work plan for the second half of the year

Jul. Dec.

• Data Expansion

Focus on expanding the faculty dataset to cover more departments and institutions, while improving profile standardization so research interests, methods, and keywords are consistently structured and easily comparable for more accurate matches.

• Quality Metrics

Introduce clear evaluation metrics for match quality such as relevance scores, engagement rates, and follow-up actions, and build a continuous user feedback loop to capture student and faculty ratings, comments, and corrections to refine recommendations.

• Web Front-End

Develop a simple, accessible web-based user interface on top of the existing terminal backend, focusing on intuitive search, match visualization, and streamlined interaction flows so students and faculty can easily explore and act on recommended connections.

• Long-Term Vision

Integrate additional scholarly data sources like publication records, grants, and collaborations, and evolve toward cross-institutional matching with analytics dashboards that help departments monitor research trends, gaps, and emerging interdisciplinary opportunities.

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Thank You

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